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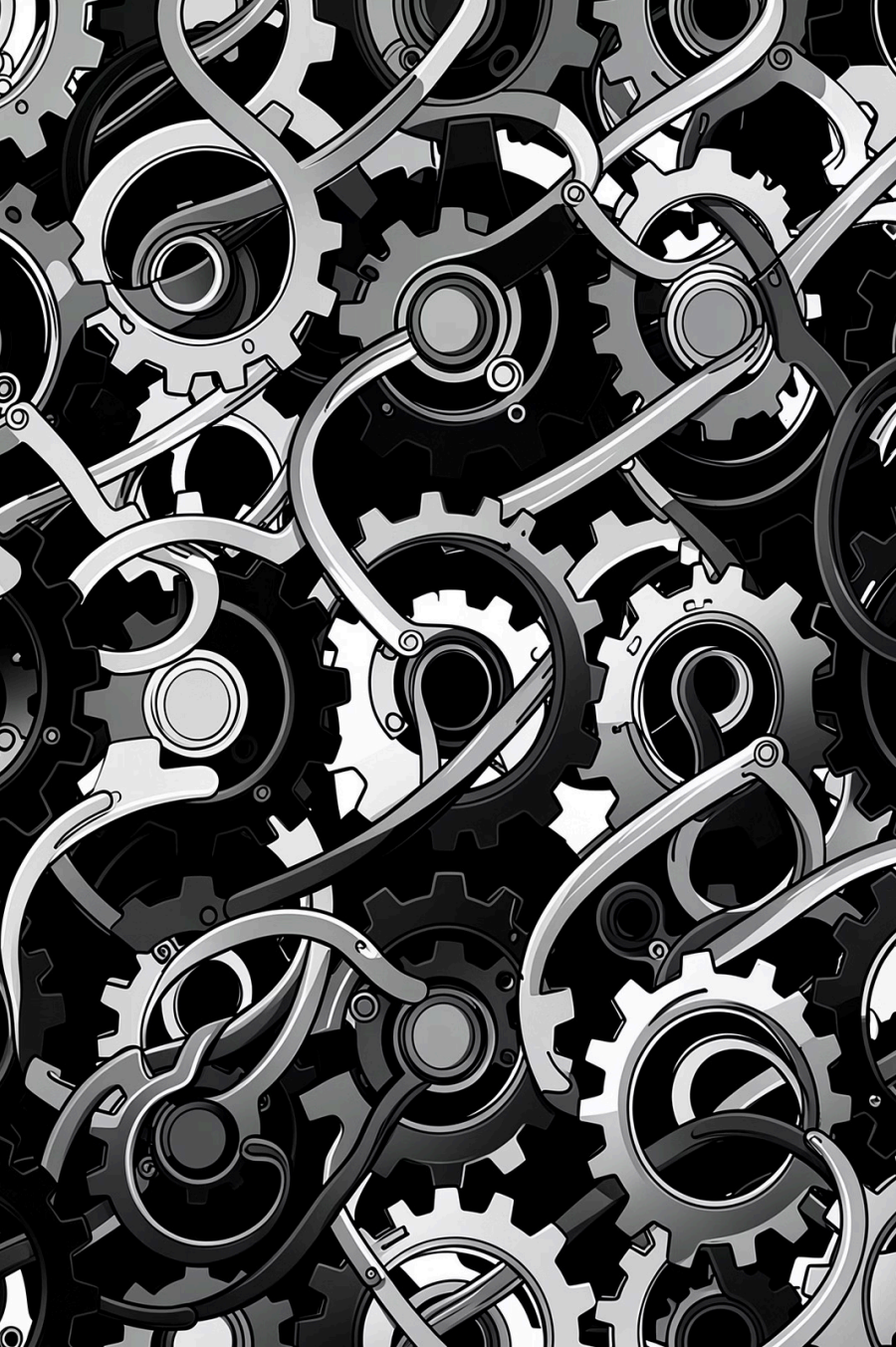
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```

For Loop in Python

Engineering Data Science (EDS) Assignment

Name: Pandurang Vyavhare | **Class:** IT1(I14) | **Roll No:** 20501080063



Introduction

Repetition is Common

In programming, repeating tasks is very common

Python Loops

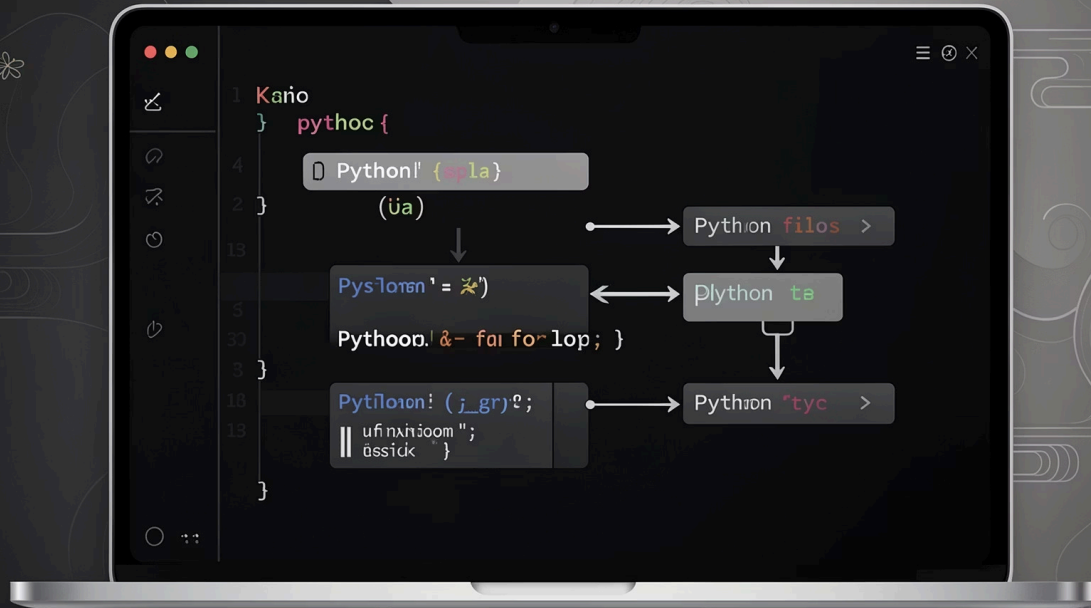
Python provides loops to handle such repetitive tasks

The For Loop

One of the most commonly used loops is the **for loop**

What is a For Loop?

A **for loop** is used to execute a block of code multiple times. It is used when the number of iterations is known. It works by iterating over a sequence (like list, range, string).



Execute Multiple Times

Runs a block of code for each item in a sequence

Known Iterations

Ideal when the number of repetitions is known beforehand

Iterates Over Sequences

Works with lists, ranges, strings, and more

Syntax & Flow

Syntax

```
for variable in sequence:  
    # block of code
```

The **variable** takes each value from the sequence one by one.
The loop runs until all values are covered.

Flow Explanation

01

Loop starts with the **first element** of the sequence

02

Code inside the loop **executes**

03

Variable updates to the **next value**

04

Process **repeats** for each element

05

Loop **stops** when the sequence ends

Basic Example & Output

Code

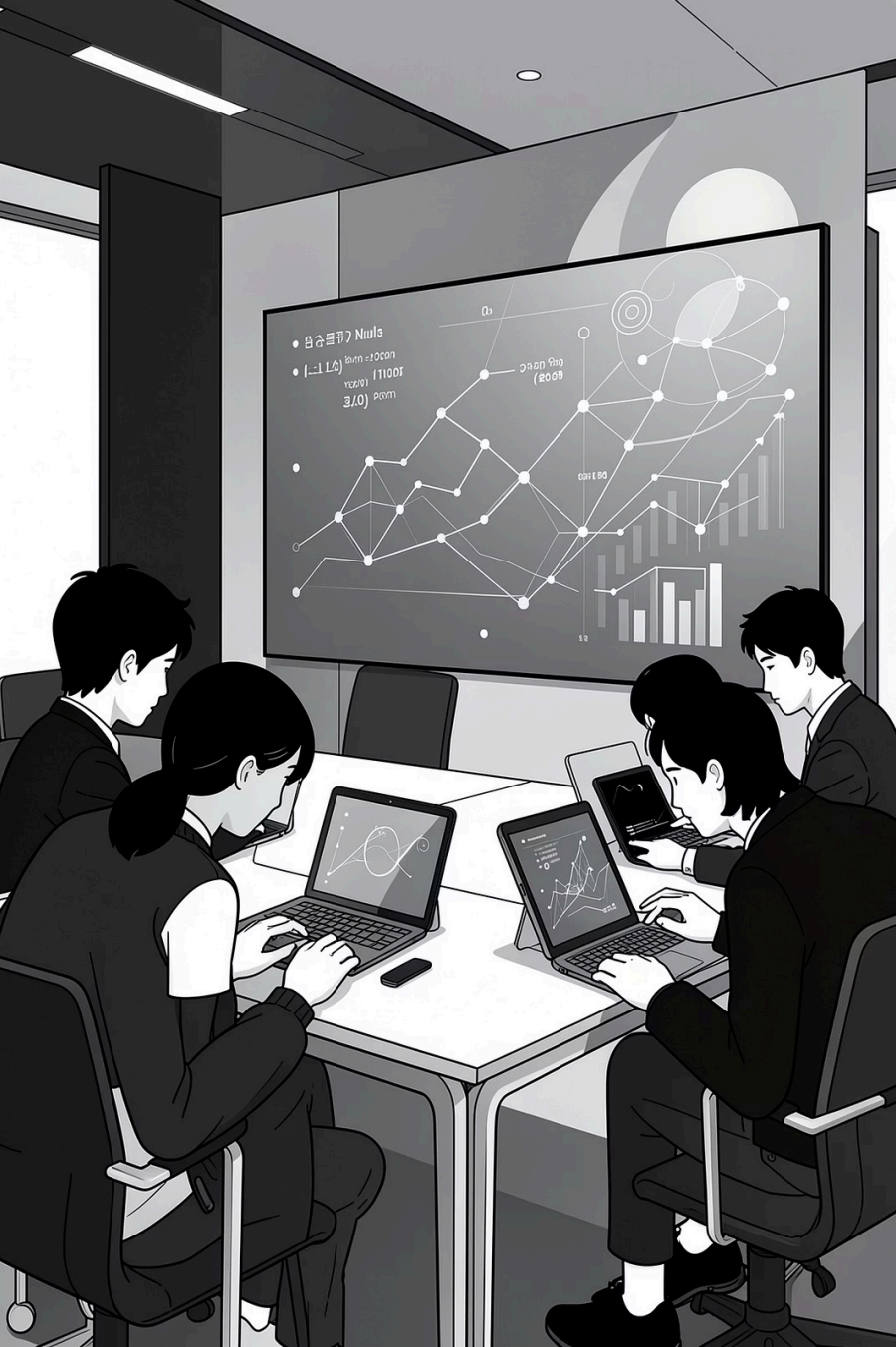
```
for i in range(1, 6):  
    print(i)
```

Explanation

- `range(1, 6)` generates numbers from **1 to 5**
- The loop prints each number one by one

Output

```
1  
2  
3  
4  
5
```



Real-Life Applications



Student Roll Numbers

Printing roll numbers for a class roster



Data Science

Iterating over datasets for analysis and processing



Automation

Repeating routine tasks without manual effort

```
students = ["A", "B", "C"]  
for s in students:  
    print("Student:", s)
```

Problem Analysis

Problem

Print the first 10 even numbers

Solution Approach

- Use `range()` with step 2
- Apply a **for loop** to iterate

Code

```
for i in range(2, 21, 2):  
    print(i)
```

Output

```
2 4 6 8 10  
12 14 16 18 20
```



Design, Conclusion & References

Design of Solution

- Identify the pattern (even numbers)
- Use correct `range()` parameters
- Apply loop efficiently
- Ensure correct output

Conclusion

The **for loop** is a fundamental concept in Python programming. It helps in automating repetitive tasks, processing data efficiently, and writing clean structured code. Understanding for loops is essential for every programmer, especially in data science and automation.

References

- Python Official Documentation
- GeeksforGeeks
- W3Schools Python Tutorial